

5th Lab

Transfer Learning & Visualizations

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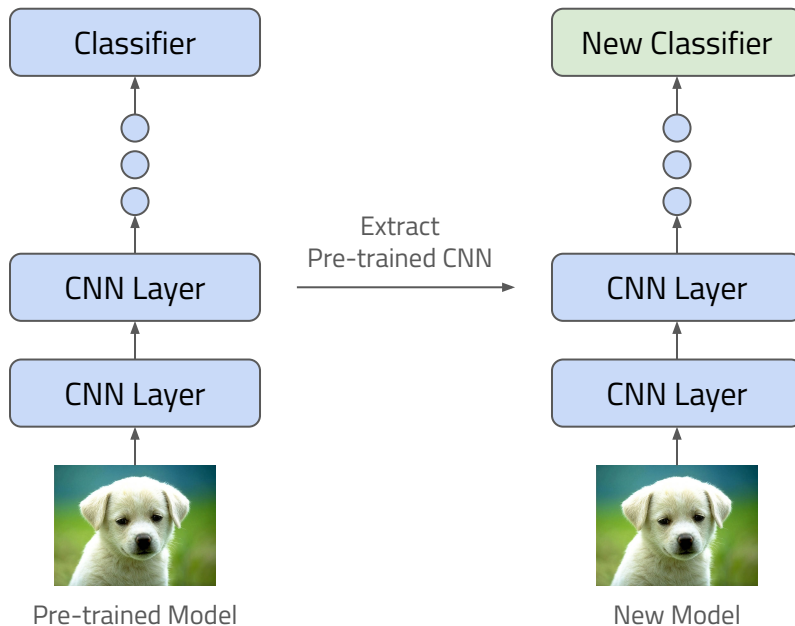


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Transfer Learning

- Most of the time, instead of creating and training a model from scratch, we can use **state-of-the-art models** (SoTA) that have been **pre-trained** on **large datasets**, such as ImageNet
- This process is called **transfer learning**



Transfer Learning

- Suppose we have a new dataset with 2 classes
- We want to **fine-tune on our dataset** the **ResNet** SoTA model, which has been trained on ImageNet

```
import torch.nn as nn  
from torchvision.models import resnet18, ResNet18_Weights
```

```
...
```

```
N CLASSES = 2
```

```
...
```

```
model = resnet18(weights=ResNet18_Weights.IMAGENET1K_V1)
```

```
in_features = model.fc.in_features
```

```
model.fc = nn.Linear(in_features=in_features, out_features=N CLASSES)
```

1) Import ReNet model and weights

2) Define number of classes to use

3) Instantiate pre-trained ResNet model

4) Get the number of input features to classification layer

5) Instantiate a new classification layer

Hands-on with the Code!

- Download the notebook on the Teaching Portal
- Upload it on Google Drive or directly on Google Colab and open it
- Download the dataset:
https://github.com/mrdbourke/pytorch-deep-learning/blob/main/data/pizza_steak_sushi.zip
- Solve the notebook
- Visualize the training loss and accuracy using TensorBoard

